UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

End of Term 1 Exam Coverage	هيكل اختبار نهاية الفصل الدراسي الأول
Grade 10 Advanced - Chemistry	**الصف العاشر - متقدم - الكيمياء**
Academic Year 2026-2027	العام الدراسي 2027-2026
Inspire	انسباير

عدد الأسئلة Number of questions	درجة السؤال Marks per question	الدرجة القصوى الممكنة Maximum overall grade
30	**(3,4)**	**100**

مدة الاختبار Exam duration	طريقة التطبيق Mode of implementation	الآلة الحاسبة Calculator
120 دقيقة **120 minutes**	**إلكتروني** **Electronic**	**يُسمح باستخدام الآلة الحاسبة العلمية** **The use of a scientific calculator is allowed**

⚠ قد تظهر الأسئلة بترتيب مختلف في الاختبار الفعلي.

⚠ Questions might appear in a different order in the actual exam.

Q	Learning outcome	Page	Exam or figure number
Q1	Differentiate between the ground and excited states of an atom	13	textbook - figure 10
Q2	Relate the principal quantum number to the total number of orbitals of the principal energy levels and the number of orbitals in sub-energy levels	22	textbook - table 2
Q3	Explain the Aufbau principle, the Pauli exclusion principle, and Hund's rule	23-24	figure 18 + table 3

Q	Learning outcome	Page	Exam or figure number
Q4	Write the electron configuration, orbital diagram, and noble gas notation of different elements	25-26	textbook - table 5
Q5	Represent the electron dot structure of atoms	28-29	table 6 + example problem 3 + practice problems
Q6	List the different families in the periodic table with the identification of the group number and the elements present	40-41	textbook
Q7	Differentiate between representative elements and transition elements	40	textbook
Q8	Use the electron configuration notation, orbital notation, and noble gas notation of an element (Z 1 - 36) to identify the location of an element in the periodic table (period, group and block)	44 - 45 - 46-48	Example 1+ figure 9+ practice problems
Q9	The electron configuration, filling sublevels	46	textbook
Q10	Explain the trend of first ionization energy across a period (Exceptions between Groups 2 & 3 and 5 & 6 are included), and down a group of the periodic table	54	textbook - figure 17
Q11	predict the periodic trend of electronegativity across a period and down a group of the periodic table	56-57	textbook figure19 +
Q12	Describe how ions (cations and anions) form to fulfill the octet rule	65	textbook

Q	Learning outcome	Page	Exam or figure number
Q13	Describe how ions (cations and anions) form to fulfill the octet rule - the pseudo noble gas configuration	66	textbook
Q14	Ion formation	66	textbook -figure 4
Q15	Describe how ions (cations and anions) form to fulfill the octet rule	69	textbook -Table 4
Q16	Explain the physical properties of ionic compounds as melting point and boiling point, conductivity when solid, molten or aqueous, and its solubility in water	72-73	textbook
Q17	Explain the physical properties of ionic compounds as melting point and boiling point, conductivity when solid, molten or aqueous, and its solubility in water	73	textbook
Q18	Write the chemical formula (using the stock method) of an ionic compound containing monoatomic and polyatomic ions	75-76-77-78	Table 7, 8 Example problem 1, 2 practice problems
Q19	Write the chemical formula (using the stock method) of an ionic compound containing monoatomic and polyatomic ions	79-80	Table 10 ,11 practice problems
Q20	Describe the formation of metallic bonds	82	textbook -figure 10

Q	Learning outcome	Page	Exam or figure number
Q21	Explain some physical properties of metals (Melting and boiling points, Thermal and electrical conductivity, Malleability, ductility, durability, Hardness and strength)	83-84	textbook -figure 11
Q22	Define, using energy diagram, covalent bond	92	textbook -figure 2
Q23	Differentiate between sigma and pi bonds	96-97	textbook+ figures 7,8,9
Q24	Identify the relationship between the type of covalent bonds (single, double and triple) and its bond length, bond strength and bond dissociation energy	98	textbook+ table1
Q25	Name a binary molecular compound based on its molecular formula	100-101	textbook + table2 Example+ problem 2 practice + problems
Q26	Determine the chemical formula of a compound from its name	102-103-104	textbook - table3- practice - problems - figure 12
Q27	Determine the chemical formula of a compound from its name	103	Table 4 practice problems
Q28	Draw Lewis structures for a number of covalent compounds with single and multiple bonds	107-108-109-110	textbook - example problems 3,4,5 practice problems

Q	Learning outcome	Page	Exam or figure number
Q29	Identify some combinations of resonance (O3 , NO3-, NO2-, SO32− and CO32−)	111-112	textbook - figure 14 - practice problems
Q30	Determine the exception of the octet rule (odd number of valence electrons, sub octets and coordinate covalent bonds, expanded octets)	112-113-114	figures 16 ,17 example problem 6 practice - problems